1. Background

Previously, we had been modeling F1 race outcomes using the Plackett-Luce (PL model. This model is a probabilistic ranking framework that directly estimates the likelihood of a full finishing order. While the PL model gave us an initial approach to model race rankings, there were several limitations.

Qualifying position as we know is a categorical and ordinal variable which in turn caused us to create dummy variables leading our total predictor count going from 6 to 25 with the addition of 19 qualifying position dummy variables. **This caused the qualifying position to capture the full prediction power of the model giving us inaccurate results and unstable coefficients, which is called quasi-complete separation.** We combatted this with adding L2 regularization and splitting qualifying positions into 3 bins (groups). This gave us stable coefficients and evaluation metrics such as Spearman of 50% and top-3 accuracy of 57%, which isn't good enough.

Plackett-Luce Limitations

The PL model **assumed a linear utility structure, which inhibited its ability to capture nonlinear interactions and the complex relationships between the predictor variables**. Along with this, the feature set was restricted to a **small number of pre-race variables** which made it difficult to capture the variation and randomness that occurs in an F1 race.

In the Plackett–Luce model, **each driver is assigned a latent utility representing their underlying competitive strength**. This utility is defined as a **linear combination of pre-race predictors and serves as the main determinant of finishing probabilities**. The model uses these utilities to construct a probability distribution over all possible race outcomes, with higher utility values corresponding to a greater likelihood of finishing ahead of other drivers.

This formula is **inherent to the Plackett–Luce framework**, as it provides a mathematically tractable way to model full rankings using a single scalar quantity per driver. However, this **structure requires that all systematic differences between drivers, which includes both performance and reliability, are captured within this single utility. Any variation not explained by the predictors is treated as random noise.**

In addition to this, the PL model relied on several assumptions. One important assumption was the **independence of irrelevant alternatives (IIA)**. This implies that the **ordering between two drivers is unaffected by the presence of other drivers**. In F1, this assumption is constantly violated due to factors like strategy, traffic, and safety cars, all introducing dependencies between multiple drivers.

While these assumptions are convenient, it is **restrictive in the context of Formula 1**, where race outcomes arise from multiple distinct processes. In particular, **reliability-related events**

such as DNFs are not purely random noise, but instead follow identifiable patterns that are influenced by different factors.

New Model Motivation

To address these limitations, a **two-staged prediction model that separated performance and reliability into two distinct components** is proposed.
In the first part, a model is used to estimate that a driver DNFs. This captures reliability-related uncertainty. The second stage is a learning to rank model that predicts the finishing order based on performance related features.
Specific model features include:

- Multi-season telemetry (2022-2025)
- Explicit modeling of driver and team reliability
- ML based ranking framework

Here, the **2022-2025 seasons are chosen**. In **Formula 1 every 3-4 years the rules and regulations change**. What this means is that the FIA (Fédération Internationale de l'Automobile), which is the **governing body of Formula 1, comes up with new rule changes in the way the car should be built.** One example of this is the current season (2026 and onwards), the rules have fully changed and power units now are to be a 50-50 power split between Internal combustion engine (ICE) and electrical power. Before it was an 80-20 split between ICE and electrical power respectively. This is the biggest change with a lot of other changes as well, but this is **done so that the performance gap reduces between teams, and to see more competitiveness.** For this reason the seasons **2022-2025 are chosen as this was the “Ground Effect Era”** where aero focus shifted away from upper-body to underbody ground effects. **Using years with different rules would cause our data to be misinterpreted and give us inaccurate predictions.**

2. Model Framework

Part 1: Reliability Modeling (DNF Probability)

A logistic regression model is used to estimate:

`P(DNF | pre-race features)` or `P(DNF | race conditions + reliability history + variability)`

The **logistic regression model estimates the probability that a driver DNFs (does not finish a race) as a function of pre-race predictors**. These predictors include both **historical reliability measures**, such as driver and team DNF rates, and **race-specific variables**, such as weather conditions, lap-time variability, and interaction effects. **Variability captures the stability of a driver’s performance, which doesn’t directly cause DNFs, but is associated with increased race risk.** The model combines these inputs through a linear function, which is then transformed into a probability using a logistic function.

This formulation allows the model to capture **how DNF risk varies across different race contexts** rather than relying solely on long-run historical averages. It accounts for the fact that **reliability is influenced not only by historical performance but also by race-specific conditions** such as weather and track characteristics. As a result, the **model estimates a context-dependent probability of failure**, reflecting structured sources of variability rather than treating DNFs as purely random events.

Along with this, logistic regression is suited for this because **DNF is a binary outcome** (finish vs. not finish), and the model directly estimates probabilities constrained between 0 and 1. Its coefficients help us understand how each factor contributes to DNF risk.

Part 2: Ranking Model (Race Outcome Prediction)

LambdaMART is a framework that learns to rank drivers by leveraging the observed finishing order in each race. It **doesn't predict the absolute prediction, it instead creates pairwise comparisons between the drivers where the true results tell us which driver should be ahead of another**. Since there are 20 drivers, this results in 190 total comparisons (n!/(r!(n-r)!) = (20*19/(2*1))), which order the relationships.
These **comparisons are evaluated based on each driver's feature vector**. This includes variables like **qualifying, team strength, driver form, and the reliability estimates** as well.
The model then **learns how the differences in these features relate to differences in finishing position**. By doing this, it recognizes patterns to determine when one driver should be ahead of another.
The model is trained to optimize ranking metrics like NDCG (Normalized Discounted Cumulative Gain), which places a greater emphasis on correctly ordering the front runners of a race.
XGBoost also has the ability to capture nonlinear relationships and interactions while having strong predictive performance.

Final Prediction

The two stages are combined as:

```
final_score = ranking_score - (2 × DNF_probability)
```

The final prediction score is constructed by combining the ranking model output along with the estimated DNF probability. The ranking model gives us a relative performance score for each driver within a race, with higher values corresponding to better expected finishing positions. To incorporate the reliability factor, a **penalty proportional to the predicted DNF probability is subtracted from this score.** The factor of **2 that is applied is a tuning parameter** controlling the relative importance of reliability versus performance. This makes sure that the driver with high failure risk is properly taken into account for the final predicted ranking. The chosen **scaling factor is heuristic** and serves as a simple mechanism to balance performance and reliability, rather than being derived from a formal statistical estimation.

3. Data and Preprocessing

We utilized FastF1 data for the 2022-2025 seasons. Each observation corresponds to one driver x one race.

3.1 Matrix

Unlike the previous Plackett–Luce model, where each race was treated as one ranked observation, the **new learning-to-rank model is built at the driver–race level**. This means each row represents one driver in one race, so the **sample size is the total number of driver–race observations, not just the number of races**. For example, one race with 20 drivers contributes about 20 rows. Across multiple seasons, this creates a much larger feature matrix while still preserving the race-level grouping structure.

This theoretically with 1 race 20 drivers equaling 20 rows. 1 season (~22 races) is 22*20 equaling 440 rows. Now 4 seasons around 90 races * 20 rows (drivers) equaling ~1800 rows. The number of races here around 88 to 92 isn't our n it is the number of groups. So in our model n ≈ 1800, groups ≈ 90, and p (predictor list) ≈ 19-20.

The model is learning to rank **within** each group. So ranking happens inside of each race (group) but the model learns across all rows

The feature matrix consists of 19 predictors for the DNF model and adds an additional feature for the ranking model. The additional feature is the predicted DNF probability.

.

3.2 Predictor Types

A key issue in the PL model was being able to handle qualifying position as a categorical and ordinal variable which created issues.

Qualifying Position

Qualifying position is both categorical and ordinal so:

- **The difference between P1 and P2 is not the same as P10 and P11**
- Treating it as continuous would introduce artificial linear assumptions

To address this, qualifying position was transformed into a percentile representation:

```
QualiPercentile = (QualiPosition - 1) / 19
```

Qualifying position is an ordinal variable, but not linearly spaced. To preserve ordering while avoiding high-dimensional dummy encoding, it is transformed into a normalized percentile. While this transformation introduces a linear scale, the use of a tree-based model allows the relationship between qualifying position and race outcome to remain nonlinear, as the model learns optimal split thresholds rather than assuming constant marginal effects.

Continuous Variables

All continuous predictors are standardized using **z-score normalization**:

$$x_{\text{scaled}} = \frac{x - \mu}{\sigma}$$

Each feature is centered by subtracting the mean and divided by its standard deviation. This ensures that all predictors are on a comparable scale. This in turn **improves numerical stability** and enables **reliable coefficient estimation** for stage 1, the **logistic regression model**. **XGBoost** is a **tree-based model**, meaning that it **doesn't require scaling**, but standardization is applied for consistency across both stages.

4. Feature Engineering

The model incorporates a comprehensive set of pre-race predictors. All predictors used in the model are either **directly observable prior to the race** (e.g., qualifying results, weather conditions, and practice session telemetry) or **derived from historical race data** (e.g., driver form, team strength, and reliability measures).

4.1 Qualifying Features

- *QualiPercentile* - normalized grid position
- *QualiGapToPole* - continuous measure of performance
- *TeammateQualiGap* - relative intra-team performance

4.2 Telemetry / Practice Features

- *FP_LongRunPace* - proxy for race pace
- *FP_LongRunVar* - consistency indicator

4.3 Performance Indicators

- *DriverForm* - rolling average of recent finishes
- *TeamStrength* - rolling team performance

4.4 Reliability Features

- *DriverDNFRate*
- *TeamDNFRate*

4.5 Environmental Features

- *AirTemp*
- *TrackTemp*
- *Rainfall*

4.6 Track characteristics

- *OvertakeIndex* - difficulty of overtaking

4.7 Interaction Terms

- *Quali x Overtake*
- *Pace x TireDeg*
- *Reliability x rain*
- *Driver_vs_Team*
    - Measures how well driver performs relative to car they have and since standardized, it becomes how many standard deviations above/below team expectation is the driver

These allow the model to capture the nonlinear effects

5. Model Selection

5.1 Logistic Regression (DNF Model)

Logistic regression is used for the DNF prediction stage. The model gives us a direct estimate of the probability that a driver fails to finish a race. This is essential in correctly integrating reliability into the final prediction.
Along with this, logistic regression allows for a clear understanding of how different features contribute to DNF risk.

5.2 XGBoost Ranking Model (LambdaMART)

As mentioned earlier this is a learning to rank framework which is well-suited for Formula 1, given that our goal is to predict the full ordering of drivers over absolute outcomes. Also, XGBoost does well in capturing nonlinear relationships and feature interactions.
While other gradient boosting methods like LightGBM or CatBoost could be used, XGBoost already gives us a stable implementation without the need for extensive feature engineering.

As mentioned above, when we do many pairwise comparisons, with 20 drivers we have

$$\binom{20}{2} = 190 \text{ comparisons}$$

If we used hypothesis testing for each pair (looking at Driver A is better than Driver B), each test has an error rate α (say 0.05). What this means is doing 190 tests then will inflate the chance for a false positive.
Formula for at least one false positive:

$$1 - (1 - \alpha)^g$$

Doing 190 tests would then cause the error to be huge, much greater than 0.05 or any alpha.
To fix this we use Bonferroni correction to shrink the per-test threshold.

$$\alpha_{\text{adjusted}} = \frac{\alpha}{g}$$

So instead of comparing each test to 0.05, we compare it to 0.05/190 or ≈0.00026. What this does is it keeps total error controlled.
The difference is in our model, XGBoost and LambdaMART don't actually use hypothesis testing. What this method actually does is optimization, and not p-values, significance thresholds, or α levels. It minimizes a loss function.
Pairwise logistic loss:

$$\log(1 + e^{-(s_i - s_j)})$$

So it optimizes metrics such as NDCG. The 190 pairwise comparisons are used as training signals, and not statistical tests. **All pairwise comparisons are jointly optimized**. The model doesn't learn if A is significantly better than B, but instead it learns to adjust scores so correct orderings are more likely overall.

5.3 Training and Evaluation

The model is evaluated using **5-fold grouped cross-validation at the race level**. The dataset is **split into five folds based on unique races**, ensuring all the drivers from a given race are assigned to the same fold. A **5-fold split was used as a standard choice to balance computational efficiency with stale and reliable estimates**.

In each iteration, approximately **80%** (depending on fold size) of races are used for training and the rest **20%** are used for testing. The process is **repeated 5 times**, with **each fold serving as the test set once**, giving a more reliable estimate of model performance.

This approach ensures that the **model is always evaluated on unseen races**. Entire races are held out from the training set, so that no information is given during training to artificially produce improved performance.

5.4 Plackett-Luce Models

5.4.2 Original PL (2024 season only)

This was the original Plackett-Luce model that I created. This model features such as *qualifying bins* (Front/Mid/Back), *DriverForm*, *TeamStrength*, *DNF Rates*, and *Overtake Index*. The problem was that the qualifying position signal was low due to it being done through bins meaning might have lost some variation due to grouping them. The features weren't efficiently engineered. Along with this we only ran the model for one season. This model exhibited low feature complexity. Our model also was trained and tested through a manually chosen training and testing percentage (70% and 30% respectively).

5.4.2 Multi-Season PL (2022-2025, 5-fold cross validation)

5.4.3 PL Enriched Feature (2022-2025, 5-cross validation)

6. Results

6.1 Feature Importance

================ FEATURE IMPORTANCE ================

	Feature	Importance
0	QualiPercentile	0.566724
13	Quali_x_Overtake	0.099013
1	QualiGapToPole	0.069611
5	DriverForm	0.065619
6	TeamStrength	0.045815
2	TeammateQualiGap	0.015663
14	Reliability_x_Rain	0.015073
15	OvertakeOpportunity	0.014736
17	GridSpread	0.014690
10	TrackTemp	0.013124
9	AirTemp	0.012759
12	OvertakeIndex	0.012564
8	TeamDNFRate	0.010080
18	DNFProbFeature	0.009868
4	FP_LongRunVar	0.009096
3	FP_LongRunPace	0.008648
16	Driver_vs_Team	0.008094
11	Rainfall	0.006927
7	DriverDNFRate	0.001895

Top Predictors:

- Qualifying position still is dominant
- Performance indicators give us a meaningful secondary signal
- Reliability is incorporated
- Interactions terms improve model flexibility

6.2 Output files

Dataset:
Metrics:
Predictions:
Feature Importance:

7. Model Evaluation Metrics

Cross-validated Metrics

Two-stage ranking model metrics

FINAL CROSS-VALIDATED METRICS	
Mean Spearman	0.653282
Mean KendallTau	0.522782
Top3 Accuracy	0.677193
Top5 Accuracy	0.749591
Winner Accuracy	0.588304
NDCG@3	0.886889
NDCG@5	0.889703
DNF LogLoss	0.681143
DNF Brier	0.243793

Enriched features PL model metrics

FINAL PL CROSS-VALIDATED METRICS	
Mean Spearman	0.664808
Mean KendallTau	0.532909
Top3 Accuracy	0.618519
Top5 Accuracy	0.756140
Winner Accuracy	0.587135
NDCG@3	0.878382
NDCG@5	0.893043
Test LogLikelihood	-718.476637
Fold	3.000000

FINAL OLD PL CROSS-VALIDATED METRICS	
Mean Spearman	0.641865
Mean KendallTau	0.510255
Top3 Accuracy	0.647953
Top5 Accuracy	0.734269
Winner Accuracy	0.532164
NDCG@3	0.873917
NDCG@5	0.882449
Test LogLikelihood	-724.666992
Fold	3.000000

Spearman

A mean spearman of about 0.54 indicates moderately strong alignment between predicted and actual rankings.

Top-3 Accuracy

A top-3 accuracy of 55% means the model identifies over half of podium drivers on average.

Top-5 Accuracy

A top-5 accuracy of 70% shows strong performance in identifying the front runners of a race.

Winner Accuracy

A winner accuracy of 53% is better but still only predicting winners accurately half of the time.

NDCG Metrics

These metrics of NDCG@3 (0.81) and NDCG@5 (0.83) show strong ranking performance for the top positions.

8. Comparison and Interpretation

8.1 Performance Comparison Overview

These results show a **clear initial growth followed by diminishing returns**

8.2 Impact of Data and Evaluation

Metric	**Original Plackett Luce (2024 season)**	**Plackett-Luce Model (2022-2025) 5-fold cv**	**Two-stage(2022-2025) 5-fold cv**
Mean Spearman	0.535	0.641	0.587
Mean KendallTau	0.350	0.508	0.450
Top-3 Accuracy	0.571	0.659	0.572
Top-5 Accuracy		0.717	0.658
Winner Accuracy		0.447	0.488
NDCG@3		0.870	0.852
NDCG@5		0.879	0.857
Feature Complexity	Low	Medium	Medium
Model Type	Linear Ranking	Linear Ranking	Nonlinear ranking

The largest improvement happens from moving from the **original single-season (2024) model to the multi-season (2022-2025), cross-validated model**. Expanding the **dataset from 23 to 92 races and along with that switching from a manually chosen single train-test split significantly stabilized the evaluation metrics**. Utilizing 5-fold CV, provides a more stable, less biased estimate of model performance by averaging results across multiple subsets.

Our original plackett-luce model **suffered from high variance due to a limited amount of data**. This made it quite **sensitive to race conditions** and variance from the 2024 season. Now, with **multi-season data and cross-validation**, the model is able to **learn more generalizable patterns** across different circuits. This shows us that the **majority of the performance gain actually comes from improved data coverage and evaluation methodology**.

8.2.1 Feature Importance/Coefficient Summary

Plackett-Luce

```
================ FINAL PL COEFFICIENT SUMMARY ================
         Feature  Coefficient  Std_Error          Z       P_Value  Abs_Coefficient
3  QualiBin_Front     0.989054   0.109883  12.736681  3.793729e-07         0.989054
4    QualiBin_Mid     0.514216   0.069226   8.891176  3.962458e-06         0.514216
6    TeamStrength     0.232394   0.060106   8.950494  1.135576e-02         0.232394
1      DriverForm    -0.206580   0.050770  -4.687306  1.068687e-03         0.206580
5    TeamDNF_Rate    -0.076126   0.076816  -1.941189  2.904298e-01         0.076126
0  DriverDNF_Rate     0.065092   0.052880   1.535514  2.100513e-01         0.065092
2   OvertakeIndex     0.000002   0.316184  -0.000003  9.999855e-01         0.000006
```

Two-Stage

```
=============== TWO-STAGE MEAN FEATURE IMPORTANCE ==
          Feature  Importance
2      DriverForm    0.353904
7    TeamStrength    0.279000
5    QualiBin_Mid    0.197568
6     TeamDNFRate    0.062135
0  DNFProbFeature    0.060104
3   OvertakeIndex    0.034643
1   DriverDNFRate    0.012645
4   QualiBin_Back    0.000000
```

<u>Interpretation:</u>

These two figures are the feature importance results for the non-enriched models. As we can see this highlights a clear difference in how each model utilizes the predictors.

In the Plackett-Luce model, **qualifying position features dominate** strongly with very low p-values and large coefficients. This tells us that the starting position of drivers is the primary predictor in race outcome with ***TeamStrength*** **and** ***DriverForm*** **giving small adjustments**.

The two-stage model however, **distributes importance across features**. *DriverForm* and *TeamStrength* actually are more influential than Qualifying predictors.

8.3 Impact of Feature Engineering (Optimal feature tuning)

Metric	**Fully enriched Plackett-Luce Model**	**Fully Enriched Two-Stage Model**
Mean Spearman	0.665	0.653
Mean KendallTau	0.533	0.523
Top-3 Accuracy	0.619	0.677
Top-5 Accuracy	0.756	0.750
Winner Accuracy	0.587	0.590
NDCG@3	0.878	0.887
NDCG@5	0.893	0.890
Feature	High	High

Complexity		
Model Type	Linear Ranking	Nonlinear ranking

The next stage presents **enhanced feature engineering**. A couple key additions include **continuous qualifying representation through *QualiPercentile*, qualifying gap features such as *QualiGapToPole* and *TeammateQualiGap***, interaction terms such as *QualixOvertake*, and contextual race features as well.
These features improve the resolution of qualifying performance and are able to **capture relationships between grid position and track characteristics**. Due to this, both models see **moderate improvements**.

8.3.1 Feature Importance/Coefficient Summary

Plackett-Luce

================ FINAL ENRICHED PL COEFFICIENT SUMMARY ================

	Feature	Coefficient	Std_Error	Z	P_Value	Abs_Coefficient
10	QualiPercentile	-2.490150	0.241384	-4.131087e+01	2.551699e-07	2.490150
11	Quali_x_Overtake	0.398934	0.252414	7.560133e+00	1.790559e-01	0.398934
15	TeamStrength	0.323451	0.204080	6.147122e+00	1.699098e-01	0.334810
8	OvertakeOpportunity	0.098202	0.251314	-5.593016e-01	3.079330e-01	0.161619
9	QualiGapToPole	0.139096	0.039000	5.157528e+00	2.806723e-02	0.139096
2	DriverForm	0.061628	0.228804	2.073141e+00	5.063142e-01	0.125799
3	Driver_vs_Team	0.078498	0.133305	1.245393e+00	2.952201e-01	0.123699
4	FP_LongRunPace	0.088990	0.069145	1.570830e+00	2.514384e-01	0.088990
16	TeammateQualiGap	-0.066493	0.034055	-2.318752e+00	8.639723e-02	0.066493
1	DriverDNFRate	0.057161	0.026973	2.133359e+00	4.882819e-02	0.057161
14	TeamDNFRate	0.029742	0.048487	7.308641e-01	5.015135e-01	0.029742
13	Reliability_x_Rain	-0.014633	0.072752	-2.759859e-01	6.911147e-01	0.028545
5	FP_LongRunVar	-0.015371	0.040808	-3.793832e-01	6.244277e-01	0.019537
6	GridSpread	-0.000014	0.866109	-1.478958e-05	9.999855e-01	0.000015
7	OvertakeIndex	-0.000011	0.997623	-1.145515e-05	9.999909e-01	0.000011
0	AirTemp	-0.000007	0.719877	-1.477280e-05	9.999864e-01	0.000009
12	Rainfall	-0.000005	1.129911	3.991849e-07	9.999927e-01	0.000008
17	TrackTemp	-0.000004	0.913120	-8.890913e-06	9.999912e-01	0.000007

Two-Stage

```
================ FEATURE IMPORTANCE ================
              Feature  Importance
0      QualiPercentile    0.566724
13     Quali_x_Overtake   0.099013
1        QualiGapToPole   0.069611
5            DriverForm   0.065619
6          TeamStrength   0.045815
2      TeammateQualiGap   0.015663
14    Reliability_x_Rain  0.015073
15  OvertakeOpportunity   0.014736
17           GridSpread   0.014690
10            TrackTemp   0.013124
9               AirTemp   0.012759
12        OvertakeIndex   0.012564
8           TeamDNFRate   0.010080
18        DNFProbFeature  0.009868
4         FP_LongRunVar   0.009096
3        FP_LongRunPace   0.008648
16        Driver_vs_Team  0.008094
11             Rainfall   0.006927
7         DriverDNFRate   0.001895
```

Interpretation:

8.4 Model Comparison

Despite the complexity of the two-stage approach, **performance gains are minimal**. The two-stage model slightly outperforms in Top-3 and winner accuracy. This suggests that this **model is more focused on identifying top positions**. On the other hand, the **PL model gives us a more consistent overall ranking performance due to its direct likelihood-based optimization**, and slightly outperforms in other metrics such as spearman, kendalltau, and top-5 accuracy.

8.5 Why Performance Plateaus

8.5.1 Limited Nonlinearity

The two-stage model is designed to **capture nonlinear interactions using XGBoost**. However, **pre-race variables such as DriverForm and TeamStrength evolve gradually**, and external factors such as weather introduce limited variation. Along with this **qualifying position related predictors capture a majority of the prediction power**.
Due to this, there is insufficient nonlinear structure for the model to exploit, leading to minimal performance gains over the simpler PL model. The PL model assumes a linear utility structure and with qualifying position being over half of the importance and a statistically significant predictor, **not being able to capture nonlinear interactions doesn't serve as a detriment to the PL model**.

8.5.2 Diminishing Returns

Once core signals such as qualifying and basic performance metrics are included, additional feature engineering yields smaller improvements. This reflects diminishing returns, where most of the predictive power has already been captured.

8.6 Differences in Model Behavior

Although the overall performance is similar, the models differ in how they make predictions.
<u>Plackett-Luce:</u>
The PL model **optimizes likelihood over full ranking**. From this it is able to provide **consistent global ordering**. This gives it slightly higher spearman and kendall.
<u>Two-Stage Model</u>
This uses LambdaMART to **optimize ranking metrics such as NDCG**. It **emphasizes the top positions** giving it a better top-3 and winner accuracy.

8.7 Main Takeaways

Based on these results there is a clear hierarchy of factors that influence model performance.

1. Evaluation methodology (train-split) and data size
2. Feature engineering
3. Model complexity

Overall, **predictive performance in pre-race F1 modeling is primarily driven by the quality of the features and the overall framework**. Once a strong set of features are established, even simpler models like PL achieve performance that is comparable to more complex ML models that only results in minimal gain.

9. Feature Correlation

9.1 Overall Feature Correlation Analysis

```
=============== FEATURE CORRELATION ANALYSIS ===============

High correlations with QualiPercentile:
Quali_x_Overtake       0.897299
OvertakeOpportunity   -0.895287
Name: QualiPercentile, dtype: float64

High correlations with DriverForm:
TeamStrength    0.822787
Name: DriverForm, dtype: float64

High correlations with TeamStrength:
DriverForm    0.822787
Name: TeamStrength, dtype: float64

High correlations with AirTemp:
TrackTemp    0.70131
Name: AirTemp, dtype: float64

High correlations with TrackTemp:
AirTemp    0.70131
Name: TrackTemp, dtype: float64

High correlations with Rainfall:
Reliability_x_Rain    0.791293
Name: Rainfall, dtype: float64

High correlations with Quali_x_Overtake:
QualiPercentile        0.897299
OvertakeOpportunity   -0.711241
Name: Quali_x_Overtake, dtype: float64

High correlations with Reliability_x_Rain:
Rainfall    0.791293
Name: Reliability_x_Rain, dtype: float64

High correlations with OvertakeOpportunity:
Quali_x_Overtake   -0.711241
QualiPercentile    -0.895287
Name: OvertakeOpportunity, dtype: float64
```

We took all predictors and computed pairwise correlations. Filtered and printed those abs(corr) >~ 0.7. This was done to identify relationships and potential redundancy between features.

Strong redundancy among engineered features

Several engineered features show high correlations. This indicates that they actually aren't introducing new information but are **just transformations of existing variables**. An example of this is *QualixOvertake* and *OvertakeOpportunity* are both highly correlated with *QualiPercentile* (≈0.90). This is expected as both of these features are directly constructed using *QualiPercentile*. These features reinforce qualifying performance rather than providing independent predictive signals which suggests there is **redundancy within the feature set**.

Driver and Team Performance Related

There also exists a strong positive correlation between *DriverForm* and *TeamStrength* (≈0.82). This reflects how F1 is structured in the real world, where **stronger teams tend to have stronger drivers**. Along with that these drivers tend to be stronger due to the car being better compared to others. While not fully redundant, these **features capture overlapping aspects of performance**, and may partially duplicate information.

Weather-related Variables Correlated
Correlations between *AirTemp* and *TrackTemp* (≈0.70) and *Rainfall* and *ReliabilityxRain* (≈0.79) show up as well. These relationships are expected due to **track temperature naturally being dependent on air temperature**, and the feature engineered variable *ReliabilityxRain* being constructed with the *Rainfall* variable. Due to this, these correlations represent feature construction rather than new meaningful relationships.
Overall
This correlation structure emphasizes that multiple engineered features, especially the interaction terms, are highly dependent on the base variables used for construction. This suggests that even **though feature engineering increases model complexity, it may not proportionally increase independent information**, mainly for qualifying related features.

9.2 Correlation with QualiPercentile

```
Correlation with QualiPercentile:
QualiPercentile        1.000000
Quali_x_Overtake       0.897299
QualiGapToPole         0.407728
TeamDNFRate            0.238234
TeammateQualiGap       0.212784
DriverDNFRate          0.090949
Reliability_x_Rain     0.054389
Rainfall               0.003697
OvertakeIndex          0.003005
GridSpread             0.001914
AirTemp                0.001239
TrackTemp             -0.000552
FP_LongRunPace        -0.012609
Driver_vs_Team        -0.134661
FP_LongRunVar         -0.172485
TeamStrength          -0.628880
DriverForm            -0.632842
OvertakeOpportunity   -0.895287
```

Fixed *QualiPercentile* variable and computed corr(QualiPercentile, feature) for all the features. This was to understand how strongly each predictor was correlated with qualifying performance.
Qualifying-derived features dominate
The strongest correlations with *QualiPercentile* are *QualixOvertake* (≈ +0.897) and *OvertakeOpportunity* (≈ -0.895). These features are effectively **just re-scaled versions of qualifying performance** and therefore **don't actually introduce new predictive information**. Instead, they increase the importance of qualifying within the model.
Moderate relationships capture additional qualifying context
Features such as *QualiGapToPole* (≈0.408) and *TeammateQualiGap* (≈0.213) both show moderate relationships. These features are **still related to qualifying but not perfectly correlated**. This means they provide additional nuance. This additional information is relative

performance within a team or field gaps. The correlations tell us that these features are more useful than purely derived features.

Driver and Team Performance Inversely Related to Qualifying

The features *DriverForm* (≈ -0.632) and *TeamStrength* (≈ -0.629) show an inverse relationship with qualifying position. What this means is that **better drivers and teams tend to qualify in stronger positions (lower percentile values)**. This then results in a negative correlation. This tells us that these features also contribute meaningful performance information beyond qualifying alone.

Independent features of Qualifying

There are a couple near zero correlations such as weather features including *AirTemp*, *TrackTemp*, and *Rainfall*, track characteristics such as *OvertakeIndex*, and race variability in *GridSpeed*. These features capture some of the key race-specific or environmental effects that aren't reflected in qualifying. This makes **these features, although weak, a potential source of independent signals**.

Overall

Qualifying performance, as expected, emerges as the dominant predictor with several **feature engineered features effectively duplicating its signal**. However, **driver and team performance metrics provide additional independent information**, while weather and track specific variables remain largely uncorrelated, capturing separate aspects of race variability.

10. Plackett-Luce Model Selection Criteria

10.1 Motivation

Unlike the **two-stage ML model, the Plackett-luce model gives a full statistical inference, including coefficient estimates, standard errors, z-scores, and p-values**. This allows for the use of formal model selection techniques that balance model fit and complexity.

Later on, a leave-one-out (LOO) method will be used for the two-stage model to evaluate predictive performance, but this method doesn't explicitly penalize unnecessary model complexity. This is why different techniques are required for the Plackett-Luce model.

10.2 Model selection methods

Several model selection techniques were considered to evaluate the contribution of the different predictors in the Plackett-Luce model.

10.2.1 Akaike Information Criterion (AIC)

AIC evaluates model quality by balancing goodness of fit with model complexity:

$$AIC = -2\log L + 2k$$

where **L is the likelihood and k is the number of parameters**. This approach encourages good model fit, allows for more flexibility in retaining predictors, and also is well-suited for predictive performance. On the other hand, it **penalizes complexity less strongly** and **may retain redundant features when predictors are highly correlated**.

10.2.2 Bayesian Information Criterion (BIC)

BIC is similar to AIC but applies a stronger penalty for model complexity:

$$BIC = -2 \log L + k \log n$$

where **n is the number of observations**. As mentioned, BIC incorporates a **stronger penalty** for unnecessary features, is **more effective at removing redundancy**, and tends to produce simpler, more interpretable models. It may remove features that provide small but meaningful improvements and can be **overly conservative** in some cases.

10.2.3 Forward Selection

Forward selection begins with an empty model and iteratively adds features that improve the selection criterion whether it be AIC or BIC. This method is simple to implement and builds the model progressively. The problem is that it **can miss important features combinations and is sensitive to correlated predictors**.

10.2.4 Backward Selection

Backward selection begins with the full model and iteratively removes the least useful features based on a chosen criterion (AIC or BIC). This starts with all the available information, is **more efficient when features are correlated**, and is well-suited when an initial full model is complete. Computationally though, this may be more expensive and may still be affected by multicollinearity.

10.2.5 Stepwise Selection (Forward-Backward)

Stepwise selection combines both forward and backward approaches by iteratively adding and removing features. This can be more flexible than purely forward or backward selection and can better navigate correlated feature spaces. Some cons include that it is more complex to implement, and is still heuristic in nature.

10.3 Method Selection

Due to the nature of the enriched feature set, which as we have come to know, contains several highly correlated predictors, model selection methods that effectively handle redundancy are preferred.

10.3.1 Primary Method: BIC + Backward Selection

This approach was chosen due to the **feature set containing substantial correlation and redundancy**, **BIC strongly penalizing unnecessary complexity**, and **backward selection being a well-suited method to refine an already fully completed model**.

10.3.2 Secondary Method: AIC + Stepwise Selection

This approach is used as a complementary method because AIC allows slightly more flexibility in retaining predictors and stepwise selection can explore a broader range of feature combinations. This serves as a robustness check against the more conservative BIC approach.

10.4 BIC + Backward Selection

10.4.1 Methodology

This process starts with the full Plackett-Luce model containing all of the predictors. Using the full dataset, the model is fit by maximizing the penalized likelihood with L2 regularization. The **log-likelihood of the fitted model is used to compute the BIC value which serves as the baseline score**.

At **each iteration, one predictor is removed at a time, and the model is refitted with the reduced predictor set**. For each refitted model, the log-likelihood is calculated and converted to a BIC score. The **predictors whose removal results in the largest decrease of BIC is permanently removed from the model**. This process is repeated iteratively until no further reduction in BIC can be achieved.

After obtaining the final set of predictors, the plackett-luce model is retrained using those features and **evaluated using 5-fold cross validation at race level**. This ensures that the selected model is assessed on its predictive performance while respecting the grouped structure of the data

10.4.2 Final Predictor set

```
Final BIC-selected predictors:
['QualiPercentile', 'TeamStrength', 'Quali_x_Overtake', 'Driver_vs_Team']
```

The BIC-backward selection method resulted in reducing the model from 18 features to 4:
QualiPercentile: **Represents the driver's normalized qualifying position relative to the field**. This as we've come to know is the strongest indicator of race outcome.
TeamStrength: **A rolling measure of team performance based on accumulated points**. This provides additional car performance and team competitiveness giving information beyond a single qualifying session. Essentially telling us how good a team's car is.
Quali_x_Overtake: **An interaction term between qualifying performance and track overtaking difficulty**. This feature gives insight on how starting position interacts with track characteristics. Poor qualifying is more detrimental on tracks where overtaking is much more difficult (eg. Monaco).
Driver_vs_Team: **This feature measures a driver's performance relative to their team's overall strength** by taking the difference between *DriverForm* and *TeamStrength*. Rather than directly comparing to their teammate, this **metric captures whether a driver is outperforming or underperforming relative to their expected performance of their car**. This gives us a more stable measure of a driver's ability across multiple races.

10.4.3 Why Features Were Removed

The remaining features were removed as they did not improve the model relative to their added complexity:
Driver Performance (*DriverForm*): While this feature does capture the momentum and recent results of a driver, it was **removed due to it being moderately correlated with both**

QualiPercentile **and** ***TeamStrength*****. Drivers that are in strong form tend to qualify well and also are on better or stronger teams, which means its predictive signal is already captured by the other variables**.
Weather Variables (*AirTemp*, *TrackTemp*, *Rainfall*): **These all showed weak and inconsistent relationships with race outcomes.**
Reliability Metrics (*DriverDNFRate*, *TeamDNFRate*, *Reliability_x_Rain*): **While conceptually important, these variables all did not provide strong predictive signals**.
Practice pace features (*FP_LongRunPace*, *FP_LongRunVar*): **These features could have introduced noise due to variability in practice conditions.**
Additional Qualifying-derived Features: (*QualiGapToPole*, *TeammateQuali*, *OvertakeOpportunity*, *GridSpread*): **These features were highly correlated with** ***QualiPercentile*** **meaning they were redundant.**

10.4.4 Coefficient Summary

```
================ FINAL ENRICHED PL COEFFICIENT SUMMARY ================
           Feature  Coefficient  Std_Error          Z       P_Value  Abs_Coefficient
1   QualiPercentile    -2.397186   0.179550 -92.420965  2.417706e-07         2.397186
3      TeamStrength     0.339908   0.070991  63.697966  3.072218e-02         0.339908
2  Quali_x_Overtake     0.306407   0.114658  29.706851  1.102885e-01         0.306407
0    Driver_vs_Team     0.097544   0.024125   5.890330  4.233112e-03         0.097544
```

The coefficient summary of the reduce model tells us the statistical significance of the new predictors:

- *QualiPercentile*: This has the the smallest p value (2.42e-7) meaning it is statistically significant, confirming that qualifying position is the dominant factor in determining race outcomes
- *Driver_vs_Team*: This feature is statistically significant (p = 4.23e-3) and captures within-team performance differences and overall driver ability relative to the car.
- *TeamStrength*: This is also statistically significant (p = 3.07e-2) and has a positive coefficient meaning that stronger teams are associated with better performance
- *Quali_x_Overtake*: This has a smaller coefficient and p-value (1.10e-1), meaning that it does contribute some interaction effect, but its impact isn't as much as the primary predictors. Along with that due to the p value it is not statistically significant.

Overall we see the reduced model was able to retain statistically significant predictors while also eliminating the redundant variables.

10.4.5 Evaluation metrics comparison

Metric	**Fully enriched Plackett-Luce Model**	**BIC + Backward Selection PL Model**
Mean Spearman	0.665	0.671
Mean	0.533	0.539

KendallTau		
Top-3 Accuracy	0.619	0.651
Top-5 Accuracy	0.756	0.767
Winner Accuracy	0.587	0.577
NDCG@3	0.878	0.882
NDCG@5	0.893	0.893

From the comparison table we see slight improvements overall (ranking performance improved slightly)**. Both Spearman and KendallTau increase, indicating better overall ranking of drivers**. **Top-3 and Top-5** accuracy **improve slightly** as well suggesting that the **reduced model is better at identifying competitive drivers near the front with reduced noise** from redundant features. On the other hand, **winner accuracy slightly decreases**, indicating a minor **loss in precision at the very top position**, which could be due to removing some fine-grained features. **NDCG remains stable**, confirming that overall ranking quality is preserved.

10.5 AIC + Stepwise

10.5.1 Final Feature Coefficient Summary

```
================ PL MODEL SELECTION: AIC STEPWISE ================
Adding QualiPercentile, AIC = 7266.661
Adding TeamStrength, AIC = 7193.846
Adding Quali_x_Overtake, AIC = 7183.134
Adding Driver_vs_Team, AIC = 7172.944
Adding DriverDNFRate, AIC = 7168.806
Adding QualiGapToPole, AIC = 7165.173
Adding TeammateQualiGap, AIC = 7162.251
No further AIC improvement.
```

```
================ FINAL ENRICHED PL COEFFICIENT SUMMARY ================
             Feature  Coefficient  Std_Error          Z   P_Value  Abs_Coefficient
3    QualiPercentile    -2.492120   0.083250 -40.120679  0.000000         2.492120
5       TeamStrength     0.369171   0.054228  11.898070  0.000421         0.369171
4   Quali_x_Overtake     0.303548   0.046767   6.863347  0.000011         0.303548
2     QualiGapToPole     0.142737   0.035676   4.884633  0.015393         0.142737
1     Driver_vs_Team     0.115688   0.023703   7.218358  0.002156         0.115688
6   TeammateQualiGap    -0.068279   0.027264  -2.661727  0.031919         0.068279
0       DriverDNFRate     0.066911   0.030956   3.419756  0.124494         0.066911
```

The AIC + stepwise selection procedure incrementally adds and removes predictors based on their contribution to model fit. Stepwise starts with no predictors and iteratively evaluates whether adding or removing features improves the AIC Score.
We see in the process the first selected feature is *QualiPercentile*, indicating this alone provides the strongest standalone predictive signal. Then additional features are added that provide additional information beyond qualifying alone.
From the coefficient summary we see again *QualiPercentile* being the dominant predictor with the largest coefficient magnitude and a highly significant p-value. *TeamStrength* and *Quali_x_Overtake* are also highly significant. Along with this, *Driver_vs_Team* is also statistically significant, suggesting that driver performance relative to team expectation gives additional exploratory power beyond raw team strength.
Overall, the AIC + stepwise model retains a larger feature set than the BIC-selected model, reflecting AIC's tendency to prioritize predictive fit over aggressive model simplification.

10.5.2 BIC + Backward vs. AIC + Stepwise Evaluation Metrics Comparison

Metric	**BIC + Backward Selection PL Model**	**AIC + Stepwise Selection PL Model**
Mean Spearman	0.671	0.667
Mean KendallTau	0.539	0.533
Top-3 Accuracy	0.651	0.629
Top-5 Accuracy	0.767	0.761
Winner Accuracy	0.577	0.577
NDCG@3	0.882	0.880
NDCG@5	0.893	0.894

From the comparison, we can see that BIC generally achieves a slightly stronger overall ranking performance for Mean Spearman and Mean Kendall, suggesting that it is more accurate in overall driver ordering. The BIC also outperforms in Top-3 and Top-5 accuracy.
The difference between the two models is relatively small, Along with this, winner accuracy is identical and NDCG@5 remains nearly unchanged with AIC slightly performing better.
The AIC method retains more predictors, allowing it to capture additional race-specific variation which resulted in the slightly improved NDCS@5, but this added complexity doesn't actually translate to consistently better predictive performance across the primary ranking metrics.

10.6 Variance Inflation Factor (VIF) Analysis For PL

10.6.1 Final Feature Set

```
Current VIF:
              Feature       VIF
12  Reliability_x_Rain  3.840116
10            Rainfall  3.628555
8              AirTemp  2.301958
9            TrackTemp  2.265754
0       QualiPercentile  2.023503
1        QualiGapToPole  1.959878
5          TeamStrength  1.936885
7           TeamDNFRate  1.596916
2      TeammateQualiGap  1.463453
3         FP_LongRunPace  1.442121
14           GridSpread  1.425630
6          DriverDNFRate  1.297172
4          FP_LongRunVar  1.221088
13        Driver_vs_Team  1.197358
11         OvertakeIndex  1.158904
```

We see that the final feature set here is the same as it was for the two-stage model. Along with that the VIF values and the feature that gets removed at each stage is the exact same. This is because VIF only depends on the feature matrix, and not on the PL or Two-Stage model. Due to this, we have the same dataset, features, and correlations so the same VIF sequence. The redundancy structure of the features is model independent.

10.6.2 VIF vs BIC Metrics

Metric	**BIC + Backward Selection PL Model**	**VIF + Backward Selection PL Model**
Mean Spearman	0.671	0.666
Mean KendallTau	0.539	0.534
Top-3 Accuracy	0.651	0.633
Top-5 Accuracy	0.767	0.756
Winner Accuracy	0.577	0.544
NDCG@3	0.882	0.875

NDCG@5	0.893	0.891

Here we clearly see that when using VIF, all the evaluation metrics performance reduces. This is expected though. BIC optimizes log-likelihood and penalty for complexity, with the main goal of the best predictive subset. On the other hand, VIF removes features that are redundant with the main goal of statistical independence. This is why in BIC, we have 4 features as it is highly optimized and best for performance. The VIF model has 15 features with less redundancy but not optimized for prediction. VIF aims to only keep features that carry independent signals not captured by other features. This means features such as weather variables (*Rainfall*, *Airtemp*, *TrackTemp*), practice features (*FP_LongRunPace*, *FP_LongRunVar*), and a couple others, although not as statistically significant, they capture prediction power that isn't explained in the other variables which is why VIF keeps them. BIC is a much stronger penalty term.

10.7 Overall

We see that the **BIC plus backward selection model achieves comparable or slightly improved performance while only using 4 instead of the original 18 predictors**. This tells us that a **majority of the feature-engineered features were redundant**, **qualifying and core performance metrics contain most of the predictive signal**, and removing **weaker features reduces noise**. These results show that a parsimonious model (model that explains data with fewer or least amount of features), based on a core set of predictors can match or outperform a highly engineered feature set.

11. Leave One Out Analysis for Two-Stage Model

11.1 Methodology

To further evaluate the contribution of individual predictors, we utilized a leave one out (LOO) feature analysis. The full model is first trained using all features to establish the baseline model performance. After, **each feature is removed one at a time and the model is retrained, and the resulting evaluation metrics are compared to the baseline**.
What this analysis does is **measure the marginal contribution of each feature to model performance**. Unlike feature importance from tree-based models, which reflects how frequently a feature is used in splits, **LOO directly looks at how model performance changes when a feature is removed**.
However, due to many of the features being highly correlated, especially the qualifying derived features, removing a single one, may not actually significantly affect results if the other correlated features can substitute for it. To address this multicollinearity, variance inflation factor (VIF) is computed for each predictor.
VIF measures **how much of a feature can be explained by other features**. For a **feature X_j, regress it on all features and compute R^2** from that regression:

$$VIF_j = \frac{1}{1 - R_j^2}$$

From our problem and the correlation analysis we already saw that *qualipercentile* is highly correlated with its engineered features and *DriverForm* and *TeamStrength* also exhibit high correlation. We can utilize iterative VIF removal, just like backward selection to find the key features.The features with high VIF are highly collinear and can be removed and after each removal the VIF values are recomputed using the new feature set.

11.2 Leave One Out Results

From this analysis, we see that **removing any individual features actually only results in minimal changes to model performance across all evaluation metrics**.
This then suggests that **many features in the model are redundant**. Qualifying related variables is an example of this as *QualiPercentile*, *QualixOvertake*, and *OvertakeOpportunity*, are all highly correlated, thus **removing one doesn't significantly reduce predictive performance as the remaining features still capture similar information**.
The results indicate that the model is robust to the removal of individual features. Many predictors, as mentioned, provide overlapping information. Single-feature importance may be understated due to multicollinearity (multiple variables are correlated, hard for the model to isolate individual effects of each predictor). Even though individual features appear to have limited impact in isolation, this doesn't imply that the information that they represent is unimportant.

11.3 Variance Inflation Factor (VIF)

11.3.1 Backward Selection Results

```
=============== VIF BACKWARD SELECTION =====

Current VIF:
               Feature          VIF
13     Quali_x_Overtake  2609.702931
15  OvertakeOpportunity  2605.633864
12        OvertakeIndex  1613.687700
5            DriverForm   356.611095
6          TeamStrength   275.608399
16       Driver_vs_Team   109.767579
0       QualiPercentile    16.068086
14    Reliability_x_Rain    3.848149
11             Rainfall     3.637078
9               AirTemp     2.305028
10            TrackTemp     2.266125
1         QualiGapToPole    1.962330
8            TeamDNFRate    1.599468
2       TeammateQualiGap    1.463769
3          FP_LongRunPace   1.443167
17           GridSpread     1.428953
7          DriverDNFRate    1.304945
4           FP_LongRunVar   1.222035
```

```
Removing Quali_x_Overtake (VIF=2609.70)

Current VIF:
               Feature          VIF
5            DriverForm   356.607443
6          TeamStrength   275.608042
15       Driver_vs_Team   109.767424
14  OvertakeOpportunity    17.938961
0       QualiPercentile    16.027931
12        OvertakeIndex     3.940469
13   Reliability_x_Rain     3.848133
11             Rainfall     3.636725
9               AirTemp     2.302154
10            TrackTemp     2.265968
1         QualiGapToPole    1.962321
8            TeamDNFRate    1.599467
2      TeammateQualiGap     1.463767
3        FP_LongRunPace     1.443154
16           GridSpread     1.428149
7         DriverDNFRate     1.304536
4         FP_LongRunVar     1.222029
```

```
Removing DriverForm (VIF=356.61)

Current VIF:
               Feature        VIF
13  OvertakeOpportunity  17.938955
0       QualiPercentile  16.023707
11        OvertakeIndex   3.940424
12   Reliability_x_Rain   3.844744
10             Rainfall   3.632136
8               AirTemp   2.302021
9             TrackTemp   2.265913
1         QualiGapToPole  1.960061
5           TeamStrength  1.937283
7            TeamDNFRate  1.596980
2       TeammateQualiGap  1.463524
3          FP_LongRunPace 1.443079
15            GridSpread  1.425673
6          DriverDNFRate  1.302160
4          FP_LongRunVar  1.221167
14        Driver_vs_Team  1.197367
```

```
Removing OvertakeOpportunity (VIF=17.94)

Current VIF:
               Feature       VIF
12  Reliability_x_Rain  3.840116
10            Rainfall  3.628555
8              AirTemp  2.301958
9            TrackTemp  2.265754
0      QualiPercentile  2.023503
1       QualiGapToPole  1.959878
5         TeamStrength  1.936885
7          TeamDNFRate  1.596916
2     TeammateQualiGap  1.463453
3       FP_LongRunPace  1.442121
14          GridSpread  1.425630
6        DriverDNFRate  1.297172
4        FP_LongRunVar  1.221088
13      Driver_vs_Team  1.197358
11       OvertakeIndex  1.158904
```

The VIF-backward selection process iteratively removes features with the highest multicollinearity until all remaining features are below the specified threshold (VIF < 5). VIF of 5 is an $R^2 = 0.8$.

At the initial stage, we see there are very high VIF values such as *Quali_x_Overtake* and *OvertakeOpportunity*, with both exceeding 2000. These variables are directly derived from *QualiPercentile* and *OvertakeIndex*, making them almost perfectly explained by other predictors. **In this first stage *Quali_x_Overtake* has a slightly higher VIF, so we remove that first.**

After removing this, the next highest VIF values are for *DriverForm* and *TeamStrength* (above 250). This is due to the strong correlation between these variables. Driver performance tends to inherently rely on team strength. ***DriverForm* is removed as *TeamStrength* provides a more stable representation of overall car performance**.

Next, we see the highest VIF being *OvertakeOpportunity*. This is removed and again it is because it is strongly related to qualifying based features. **After this removal we see that all of the remaining features exhibit VIF values below the threshold of 5**.

11.3.2 New Feature List

The remaining features are:

- *QualiPercentile*, *QualiGapToPole*, *TeammateQualiGap*: These all capture different aspects of qualifying performance such as relative position, gap to pole (leader), and intra-team comparison
- *TeamStrength* and *Driver_vs_Team*: These represent team performance and and driver performance relative to their teammate, this allows for the model to separate car and driver effects
- *FP_LongRunPace*, *FP_LongRunVar*: These are practice session metrics that give insight in race pace and consistency

- *DriverDNFRate*, *TeamDNFRate*, *Reliability_x_Rain*: These are all reliability-related features that capture the likelihood of race incidents and failures
- *AirTemp*, *TrackTemp*, *Rainfall*: These are environmental variables that may influence how they race plays out
- *OvertakeIndex* and *GridSpeed*: These are track-level characteristics and grid variability that affect overtaking and race competitiveness

Why These Remain:

BIC selection strongly penalizes multicollinearity and reduces the model to a minimal subset. **VIF-based selection removes features that are statistically redundant**. The **retained features each give specific information** that can't be sufficiently explained by the others, **even if their direct predictive contributions are small**.

11.3.3 Evaluation Metrics Comparison

Metric	**Fully Enriched Two-stage Model**	**VIF Reduced Two-Stage Model**
Mean Spearman	0.653	0.657
Mean KendallTau	0.523	0.522
Top-3 Accuracy	0.677	0.688
Top-5 Accuracy	0.750	0.758
Winner Accuracy	0.590	0.588
NDCG@3	0.887	0.890
NDCG@5	0.890	0.892

We see that **overall ranking performance is preserved**. Spearman and KendallTau remain nearly identical, showing that the relative ordering of drivers is unchanged. Top-3 and top-5 accuracy improve slightly showing that removing redundant features potentially reduces noise. Winner accuracy is stable and the slight decrease doesn't indicate a meaningful loss in predictive power. Along with this NDCG improves slightly confirming that overall ranking quality is maintained. The results demonstrate that **removing multicollinearity does not harm predictive performance**. Instead, it **slightly improves generalization by reducing redundancy** and stabilizing the model.

11.4 Overall

Overall, the leave one out analysis reveals that while the enriched feature set may contain many engineered predictors, **predictive performance is primarily driven by a small subset of core**

information. The leave-one-out (LOO) analysis shows that **removing individual predictors results in only minor changes in performance**. This indicates a **high degree of redundancy** among features. This is also supported in the VIF analysis, which identifies significant multicollinearity.

The **VIF-based backward selection confirms that many predictors are strongly explained by others**, and their **removal actually does not meaningfully impact model performance**. Despite reducing multicollinearity, the VIF-reduced model still retains a large feature set, emphasizing **while features may be correlated, they still capture unique aspects of race dynamics**.

Across both the VIF and LOO analysis, we see **qualifying performance still being the dominant driver of predictive accuracy, with team strength and driver-related metrics providing additional but smaller contributions**. These are also consistent with the correlation analysis done prior, which showed strong relationships among engineered features.

Overall, the results indicate that increasing feature complexity doesn’t necessarily improve predictive performance. Carefully selecting features that provide independent information leads to more stable models without sacrificing accuracy.

12. Conclusion